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Introduction to Deep Learning

Introduction and Overview



Acknowledgements

Slides of this lecture make use of / are based on:

- Andreas Geiger (University of Tübingen): Deep Learning (https://uni-tuebingen.de/fakultaeten/mathematisch-naturwissenschaftliche-fakultaet/fachbereiche/informatik/lehrstuehle/autonomous-vision/lectures/deep-learning/)
  Here: slide deck 1 (Introduction)

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1.1 Welcome



Welcome from our guests!

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https://tryparrotai.com/video?id=w_xoafh2g2

https://tryparrotai.com/video?id=w_ax1yol2z

Nobel Prizes 2024

The Nobel Prize in Physics 2024

John Hopfield

"for foundational discoveries and inventions that enable machine learning with artificial neural networks"



John Hopfield. Ill. Niklas Elmehed © Nobel Prize Outreach

Geoffrey Hinton

"for foundational discoveries and inventions that enable machine learning with artificial neural networks"



Geoffrey Hinton. Ill. Niklas Elmehed © Nobel Prize Outreach

https://www.nobelprize.org/all-nobel-prizes-2024/

Nobel Prizes 2024

The Nobel Prize in Chemistry 2024

David Baker

"for computational protein design"



David Baker. Ill. Niklas Elmehed © Nobel Prize Outreach

Demis Hassabis

"for protein structure prediction"



Demis Hassabis. Ill. Niklas Elmehed © Nobel Prize Outreach

John Jumper

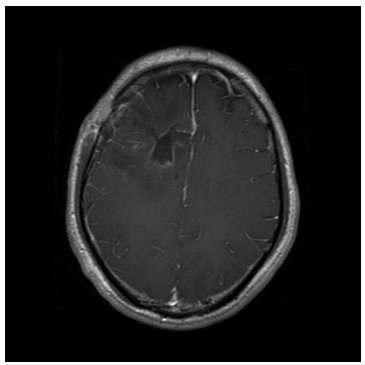
"for protein structure prediction"



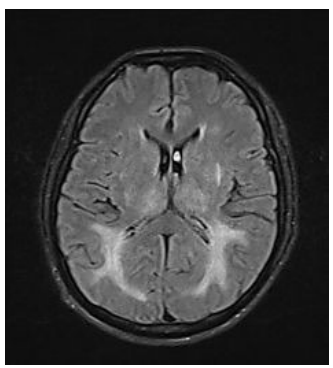
John Jumper. Ill. Niklas Elmehed © Nobel Prize Outreach

https://www.nobelprize.org/all-nobel-prizes-2024/

Medical Image Classification



Glioma
(Tumor in the central nervous system)



No Tumor

Fraud Detection

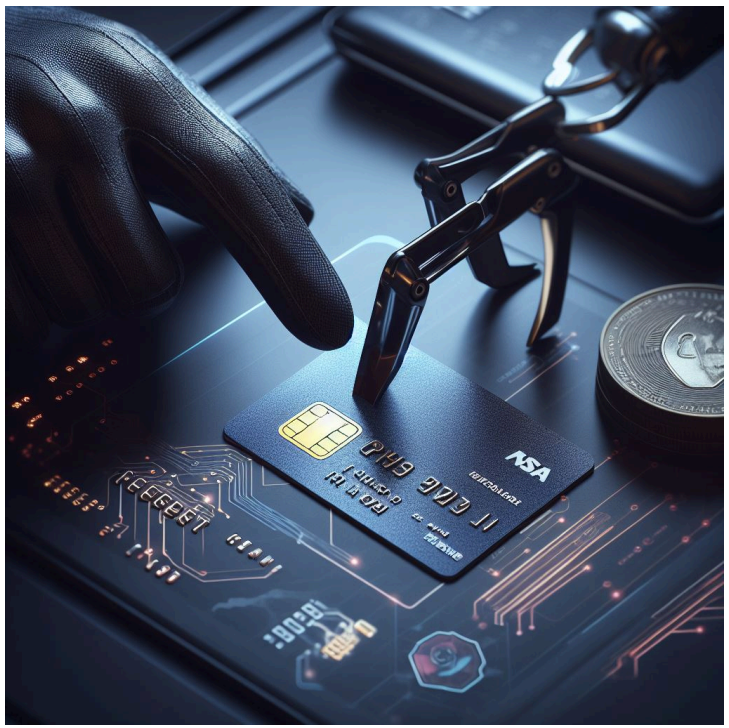


Image by Dall-E 3 via Bing Image Generator

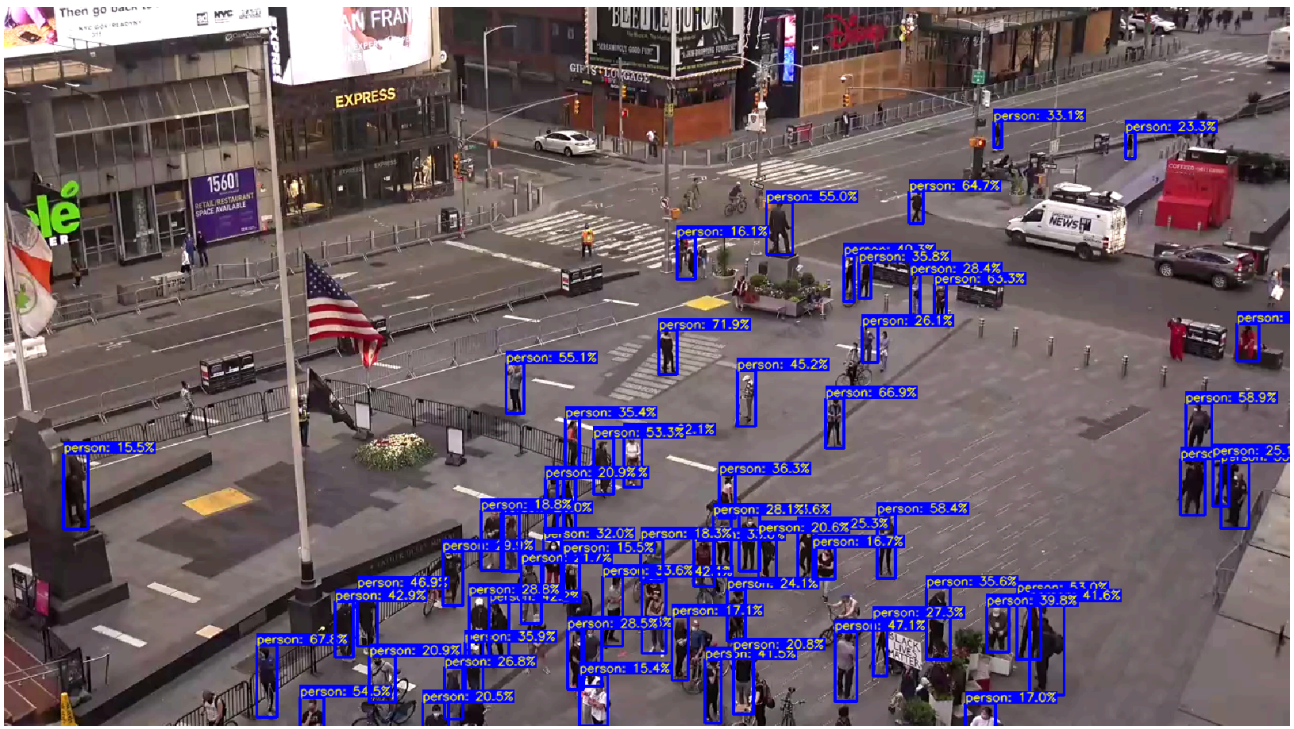
Speech Recognition



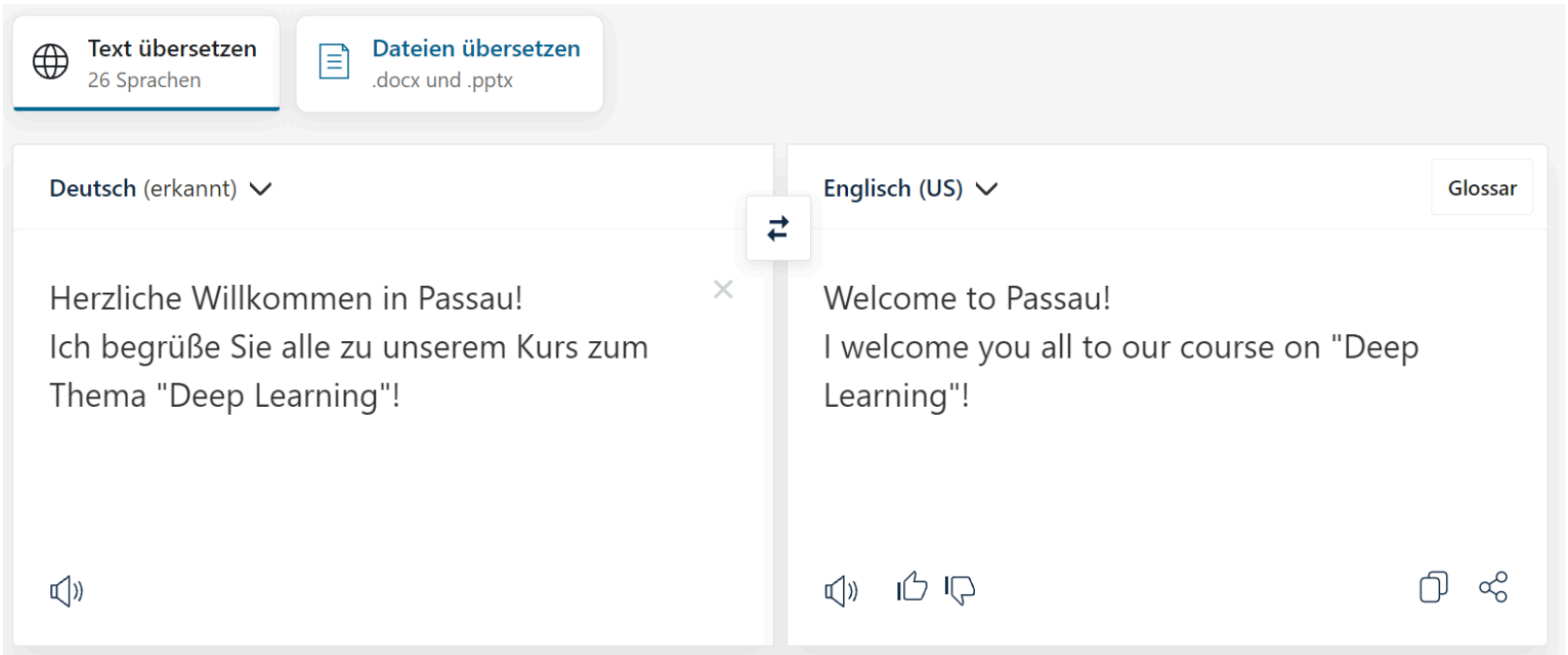
https://www.amazon.de/echo-dot-2022/dp/B09B8X9RGM

Image by Dall-E 3

Object Detection

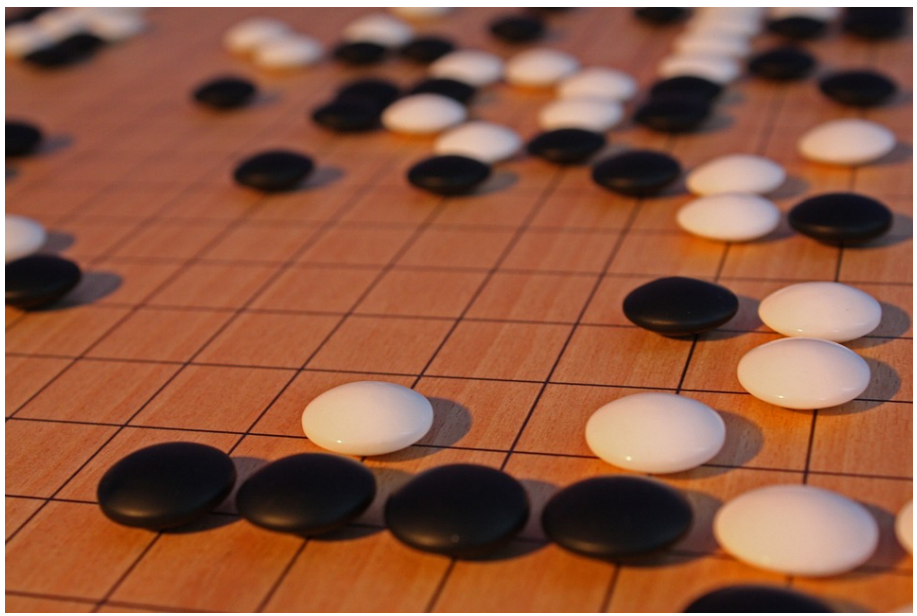


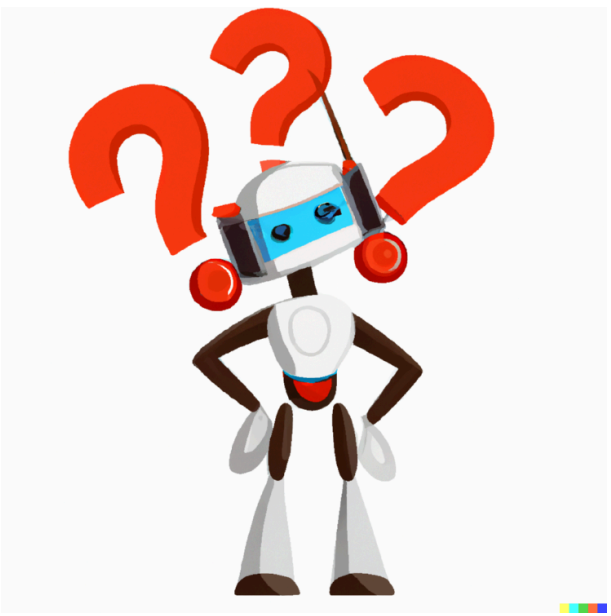
Translation



Screenshot from translate.google.com

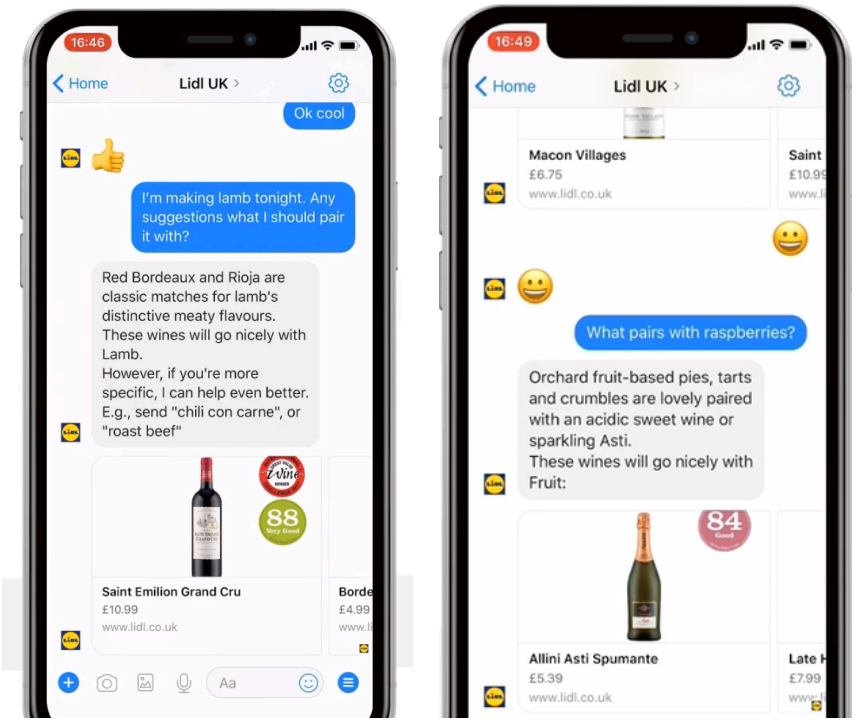
Deep Learning in Games





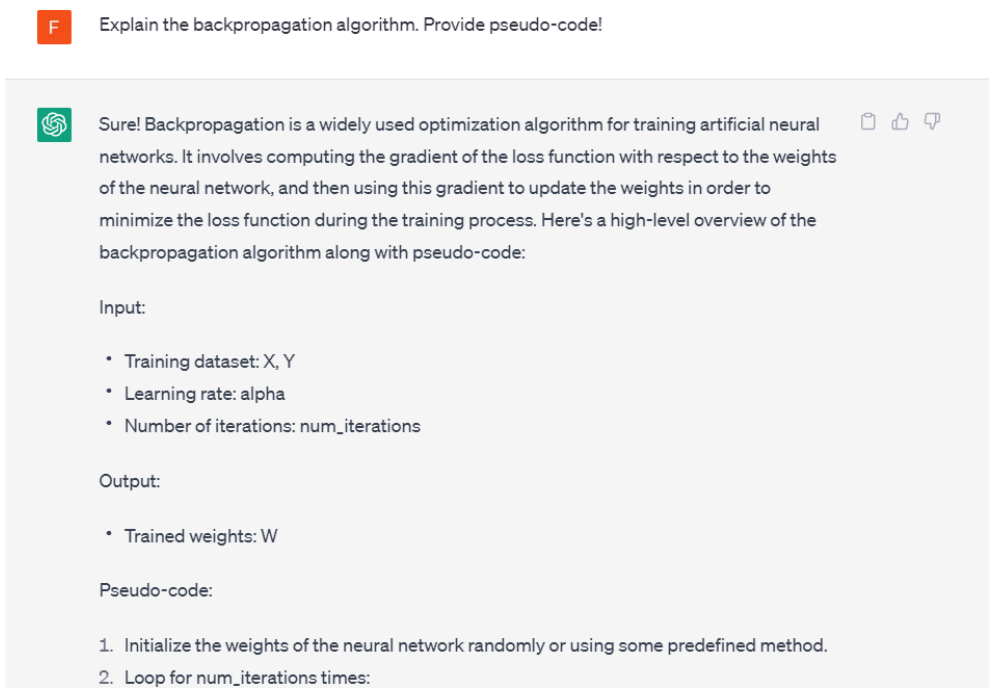
Dall-E 2

Chat Bots 2021



https://www.chatbotguide.org/lidl-chatbot

Chat Bots 2023



F Explain the backpropagation algorithm. Provide pseudo-code!

Sure! Backpropagation is a widely used optimization algorithm for training artificial neural networks. It involves computing the gradient of the loss function with respect to the weights of the neural network, and then using this gradient to update the weights in order to minimize the loss function during the training process. Here's a high-level overview of the backpropagation algorithm along with pseudo-code:

Input:

- Training dataset: X, Y
- Learning rate: alpha
- Number of iterations: num_iterations

Output:

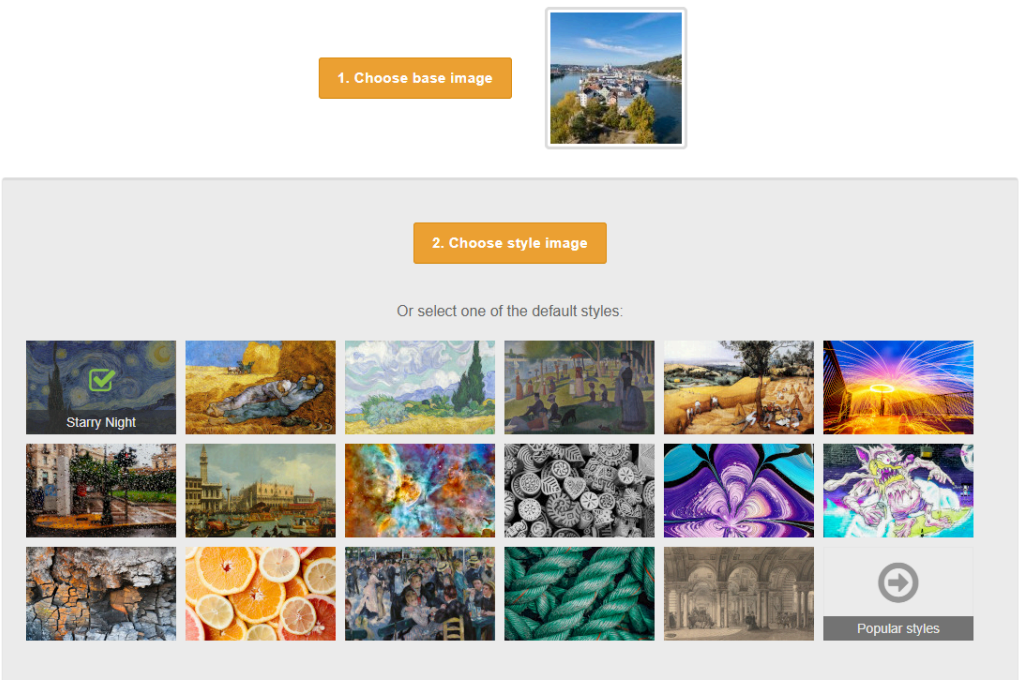
- Trained weights: W

Pseudo-code:

1. Initialize the weights of the neural network randomly or using some predefined method.
2. Loop for num_iterations times:

Screenshot from https://chat.openai.com/

Image Generation 2022

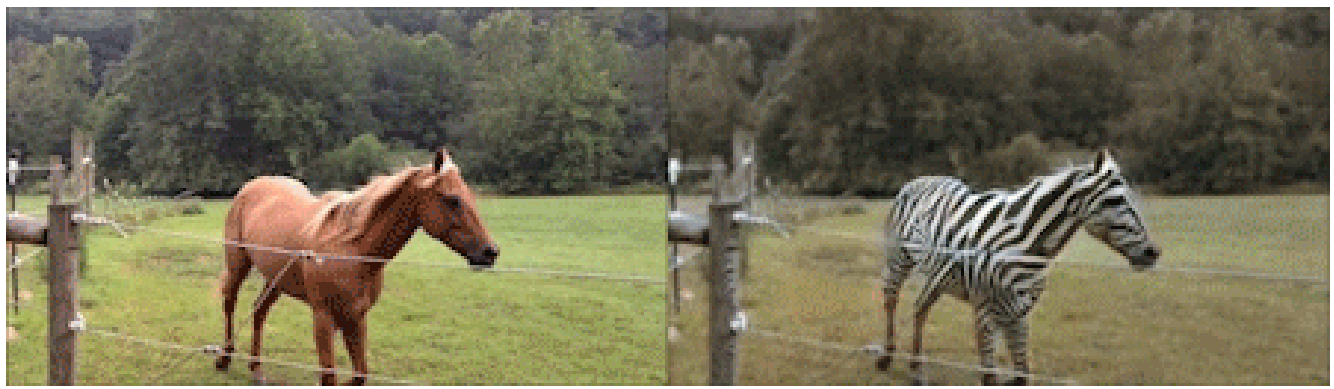


https://deepdreamgenerator.com/generator

Image Generation: Result



Image-to-Image Translation



https://github.com/junyanz/CycleGAN

More applications

- Link prediction in social networks
- Credit card fraud detection
- Gender detection from text
- News aggregators (also with sentiment)
- AI that plays arcade games
- Image caption generation
- Grammar checker
- Inventing new algorithms(!)

Agenda



- Welcome & Motivation
- Examples of Deep Learning
- Course Organization
- What is Deep Learning?
- History of Deep Learning

Prof. Florian Lemmerich

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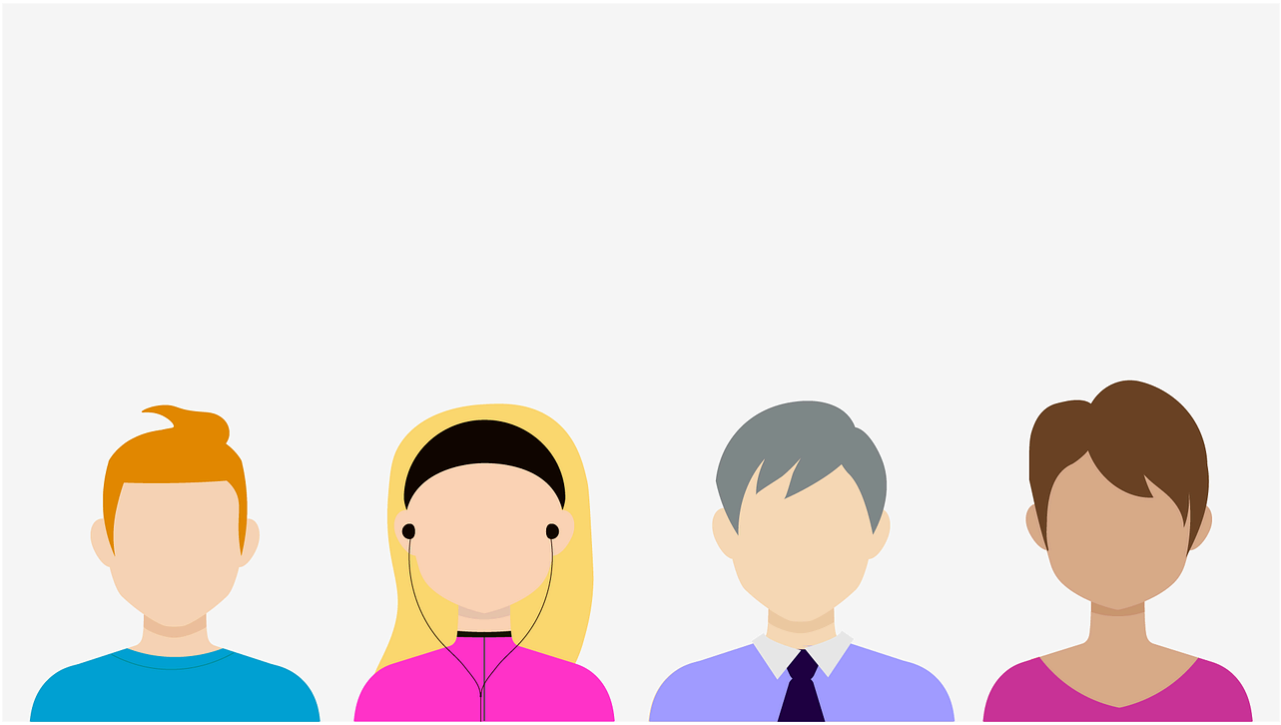
- Studied in Würzburg (Computer Science, Math, History)
- PhD in Computer Science (Data Mining) in Würzburg
- Postdoc at “GESIS – Leibniz Institute for the Social Sciences”, Cologne
- “Senior Postdoc” at RWTH 2017-2021
- Professor for Applied Machine Learning at Passau since April 2021

Andreas Einwiller, M.Sc.

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- M.Sc. 2025
- Teaching assistant for the exercises

Who are you?



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1.2 Course Organization



Course Goals

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- Understand the **fundamentals** of Deep Learning
- Know the basic concepts, terminology, algorithms
  → enables you to look up and understand new content quickly
- Be able to apply Deep Learning in your own small projects
  → including some implementation (in the exercises)
- Give a glimpse into current research
- …but we are only scratching the surface

Course dates and attendance

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- Lectures: Friday 10.15h – 11.45h, (WIWI) HS 8
  16.10.2026 – 05.02.2027
- Exercises:
  - Master students: Thu 12.15h – 13.45h or Thu 14.15h – 15.45h in (WIWI) SR 029
  - Bachelor students: Tue 14.00h – 14.45h
- Language: English
- You should attend both lectures and exercises to succeed in this course!
- Attendance is not mandatory, but highly recommended
- Course slides will be put online in StudIP
- No intentional omissions, but completeness of the slides is not guaranteed
- “The spoken word counts” (for exam content)
  ➔ You should take notes

Interactions

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- There is a lot of (excellent) material on this topic

- Advantage of this course: Interactivity!
- If something is unclear at the moment: ask immediately
- Additional opportunity to ask questions after each section

- Please use the forums to ask questions and help answer them

- (Almost) no question is a stupid question!

Why no lecture recording?

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- Course under development, some parts are not yet in the shape I would like
- Experience:
  Fewer questions, higher threshold for interaction when lectures are recorded
- Allows speaking “off the record”
- Please do not record privately

Course Overview (tentative)

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Session	Date	Topic
1	16.10.2026	Introduction
2	23.10.2026	Foundations of Machine Learning
3	30.10.2026	Foundations of Machine Learning
4	06.11.2026	Single-Layer Neural Networks
5	13.11.2026	Single-Layer Neural Networks
6	20.11.2026	Multi-Layer Neural Networks I
7	27.11.2026	Multi-Layer Neural Networks II
8	04.12.2026	Multi-Layer Neural Networks III – Regularization
9	11.12.2026	Convolutional Neural Networks
10	18.12.2026	Convolutional Neural Networks
11	08.01.2027	Recurrent Neural Networks
12	15.01.2027	Language Models
13	22.01.2027	Transformer Models
14	29.01.2027	Generative Models
15	05.02.2027	Explainability & Wrap-up

Exams

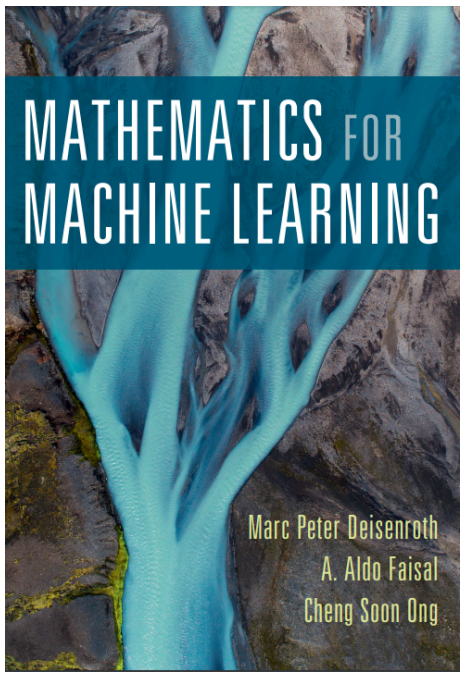
- The exam will be a written exam of 90 minutes
- No prerequisites to participate (e.g. exercise scores)
- No bonus
- Do not forget to register in time!
- Covers everything covered in the course:
  - Lectures and exercises
  - May contain some coding
  - Including first (!) and last (!) lecture
- There will be a second exam (resit) before the beginning of the next semester
- Dates for both exams will be announced as soon as possible

Assumed Knowledge

- Linear Algebra, including:
  - Vectors, matrices
  - Vector and matrix operations (e.g., multiplication)
  - Matrices as functions
- Statistics
  - Data types
  - (Conditional, joint) probability, likelihood
  - Distributions

- Calculus
  - Minima, Maxima
  - Derivatives, differentiation rules
  - Partial derivatives & gradients

- Recommendation:
  Deisenroth, Faisal, Ong:
  Mathematics for Machine Learning
  (free PDF: mml-book.github.io)



Material: online courses

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- Many (many, many) courses out there
- Lecture: Deep Learning (Prof. Andreas Geiger, University of Tübingen)
  https://www.youtube.com/playlist?list=PL05umP7R6ij3NTWIdtMbfvX7Z-4WEXRqD

  ➔ Closest to our course, material used from there

Material: Books

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- Goodfellow, Bengio, Courville: Deep Learning
  https://www.deeplearningbook.org/
- Aggarwal, Charu: Neural Networks and Deep Learning
  http://www.charuaggarwal.net/neural.htm
- Zhang, Lipton, Li, Smola: Dive into Deep Learning
  http://d2l.ai/

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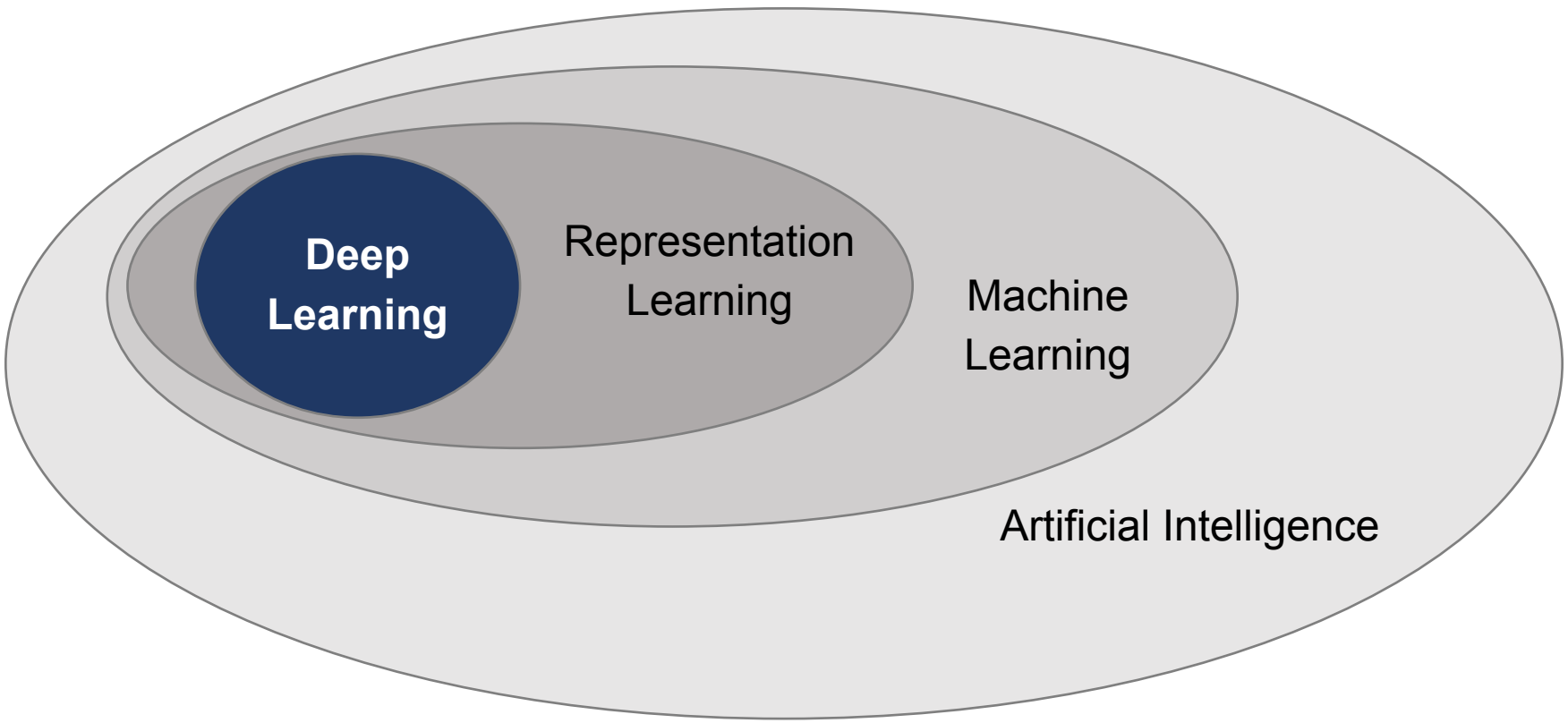
1.3 What is Deep Learning?



What is Deep Learning?

- Deep Learning is a buzzword!
- The term “deep learning” was/is a “marketing trick”
- Formerly just known as “multilayer artificial neural networks” or “multilayer perceptrons”

Deep Learning

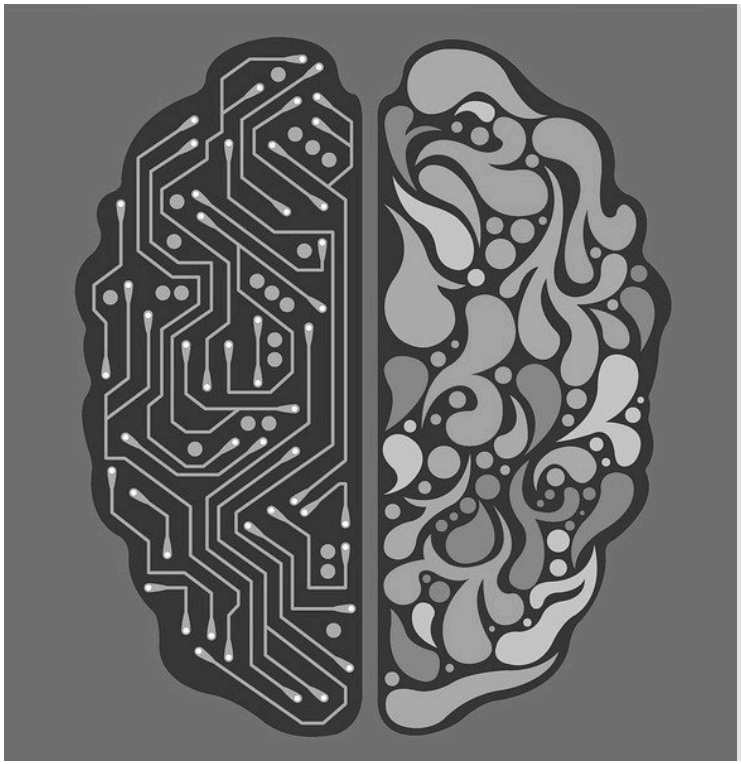


Artificial Intelligence

- Artificial Intelligence:

  “Intelligence exhibited by machines or software”

Artificial Intelligence (AI)



- ***"Intelligence exhibited by machines or software"***
- ***Intelligence:***
  - *"a general capacity of an individual consciously to adjust his thinking to new requirements" (Stern 1914)*
  - *"Intelligence is, what an intelligence test measures" (Boring 1912)*
- *"AI is whatever computers cannot do yet"* (moving goal posts)

Subfields of Artificial Intelligence

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- Logic and Reasoning
- Knowledge Representation
- Automated Planning and Scheduling

- **Machine Learning**

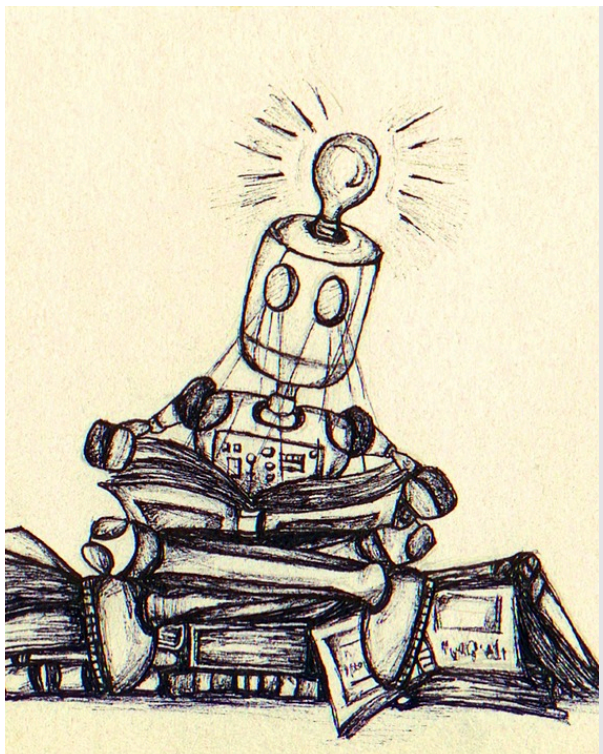
- Examples:
  - Solving constraint problems
  - Applying manually created rules
  - …

AI and Deep Learning

> *"Realistically, deep learning is only part of the larger challenge of building intelligent machines. Such techniques lack ways of representing causal relationship (…) have no obvious ways of performing logical inferences, and they are also still a long way from integrating abstract knowledge, such as information about what objects are, what they are for, and how they are typically used. The most powerful AI systems, like Watson (…) use techniques like deep learning as just one element in a very complicated ensemble of techniques, ranging from the statistical technique of Bayesian inference to deductive reasoning."*
>
> – Gary Marcus

Zhang et al.: On Definition of Deep Learning
https://ieeexplore.ieee.org/abstract/document/8430387

Machine Learning (ML)



- *"**Machine learning** (ML) is the study of computer algorithms that can improve automatically through experience and by the use of data. It is seen as a part of artificial intelligence. Machine learning algorithms build a model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so." –* Wikipedia
- *Learn a **model** from **training** examples,*
  *apply model to make **predictions** about the future*

Machine Learning Tasks

- Classification
- Regression (numeric prediction)
- Structured Prediction

- Clustering

- Reinforcement Learning
- …

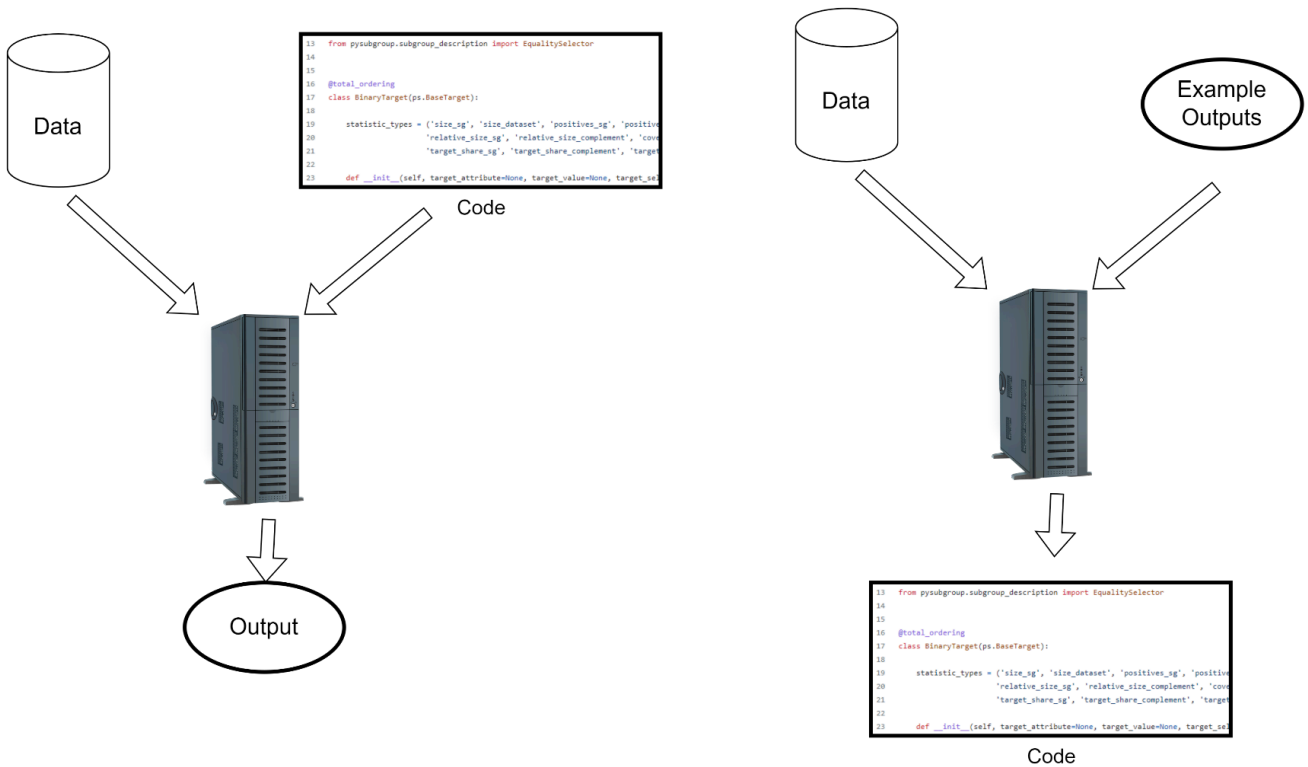
Deep Learning and Machine Learning

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> *"Deep Learning is a subfield of machine learning concerned with algorithms inspired by the structure and function of the brain called artificial neural networks."*
>
> – Jason Brownlee, machinelearningmastery.com (2016)

https://machinelearningmastery.com/what-is-deep-learning/

Also cited in: Zhang, W. J., et al. "On definition of deep learning." *2018 World automation congress (WAC).* IEEE, 2018.


Data
Code
Output
Data
Example Outputs
Code

Teaser

- More on Machine Learning: Next Week

Representation Learning

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- Machine learning typically uses *features* (properties) of data instances to use in predictions
- Good features are often not directly contained in the data

- Traditionally, features were crafted by hand to help prediction
- Features can also be created automatically, e.g., by identifying rules

Representation Learning: Definition

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> *"[…] use machine learning to discover not only the mapping from representation to output but also the representation itself. This approach is known as* ***representation learning****."*
>
> – Goodfellow et al. (2016)

> *"[…] learning representations of the data that make it easier to extract useful information when building classifiers or other predictors."*
>
> – Bengio et al. (2013)

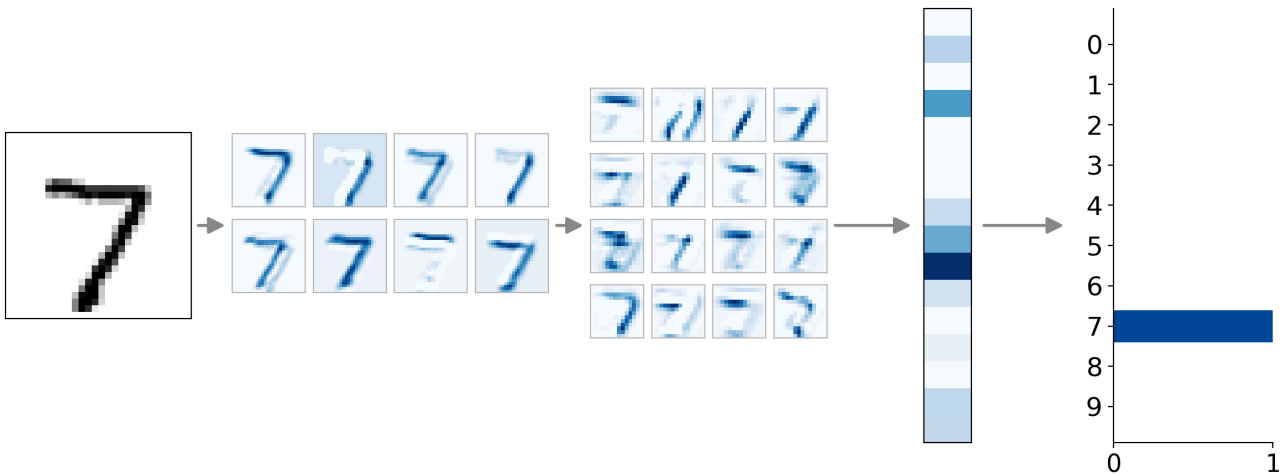
$$\boldsymbol{x} \xrightarrow{\varphi} \boldsymbol{h} = \varphi(\boldsymbol{x}) \xrightarrow{g} \hat{y} = g(\varphi(\boldsymbol{x}))$$

Learn *both* g and the representation φ from data

Goodfellow, Bengio, Courville: *Deep Learning*, MIT Press, 2016, Ch. 1

Bengio, Courville, Vincent: Representation Learning: A Review and New Perspectives. *IEEE TPAMI* 35(8), 2013

Example: Learned Representations



- The *same* digit, represented differently in each layer
- No feature was designed by hand: all were learned from 60,000 training images

Own example: small CNN trained on MNIST (2 epochs, 96% test accuracy)

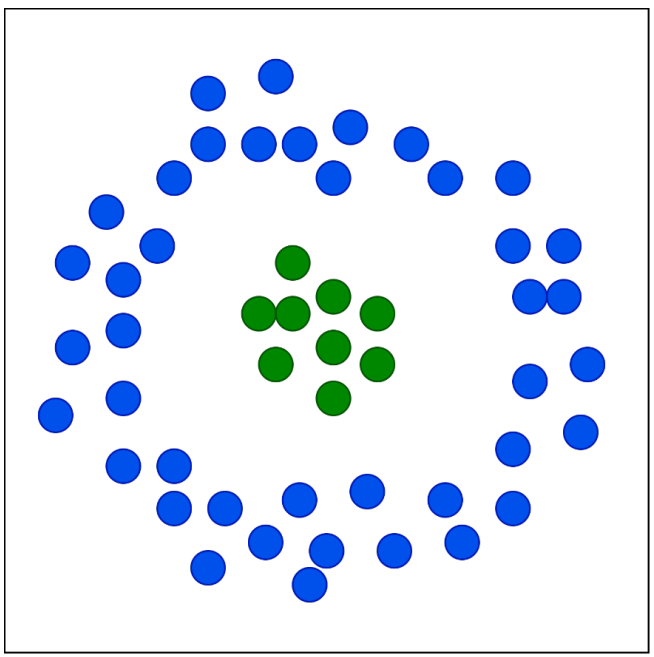
Example: Representations



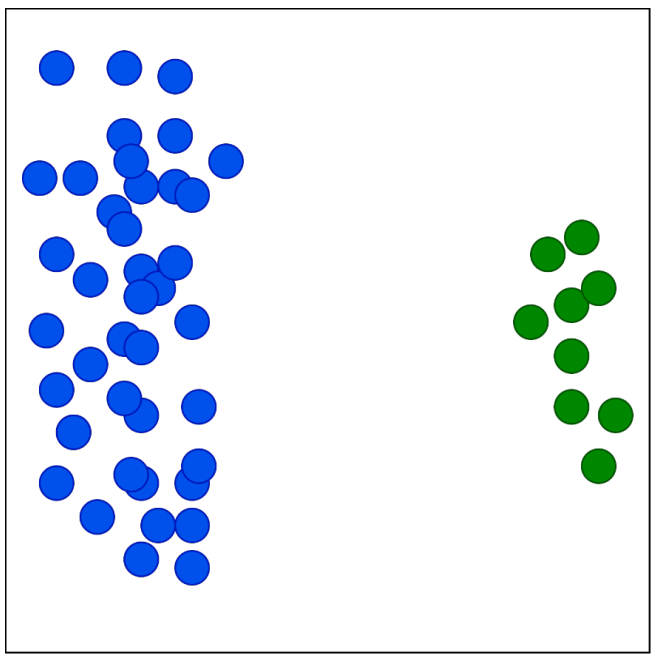
From: “Deep Learning - A Visual Approach” by Andrew Glassner

Representation matters

Cartesian coordinates

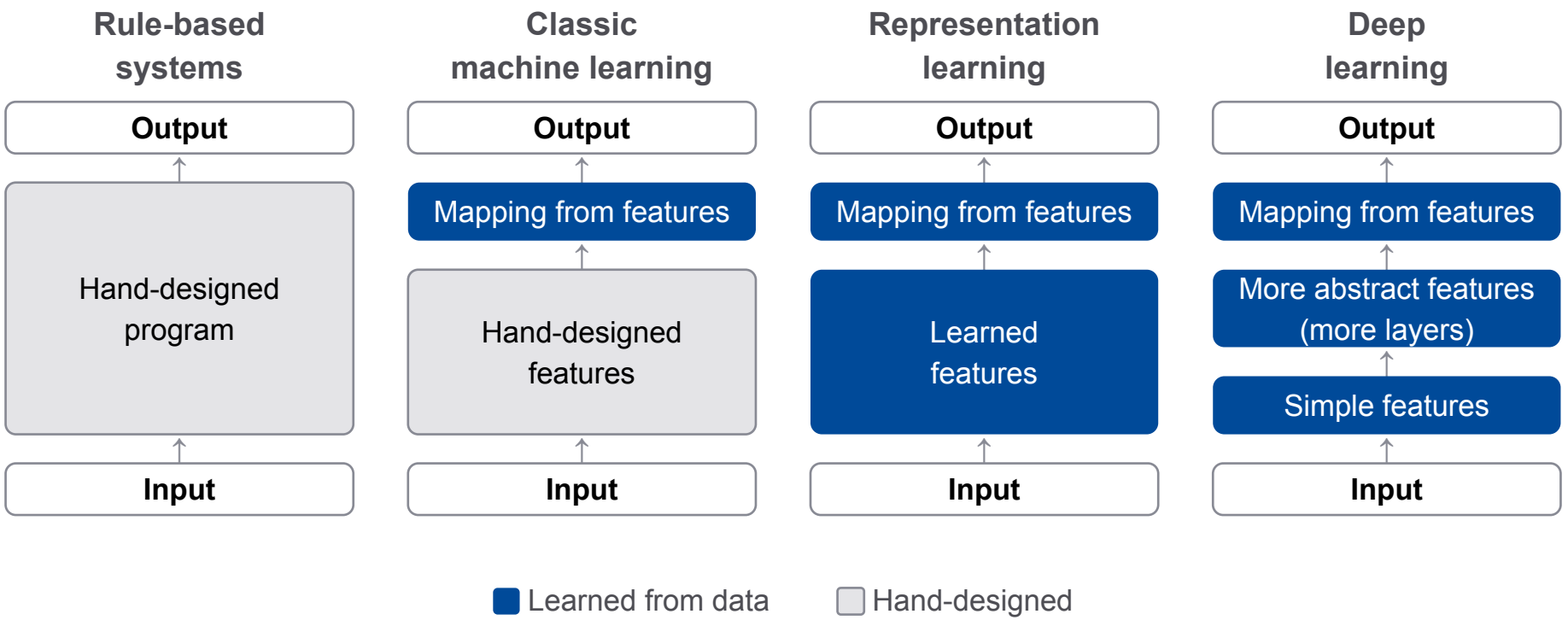


Polar coordinates



Slide idea: https://www.deeplearningbook.org/slides/01_intro.pdf

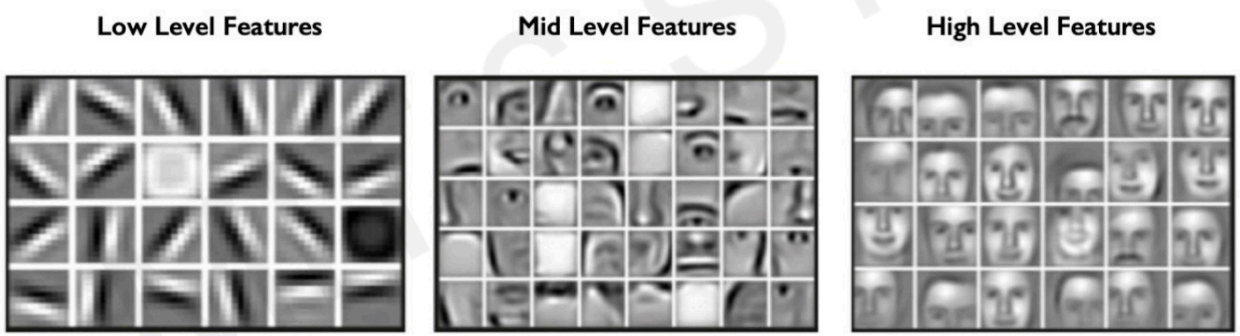
Where Does Learning Happen?



Adapted from: Goodfellow, Bengio, Courville: *Deep Learning*, MIT Press, 2016, Fig. 1.5

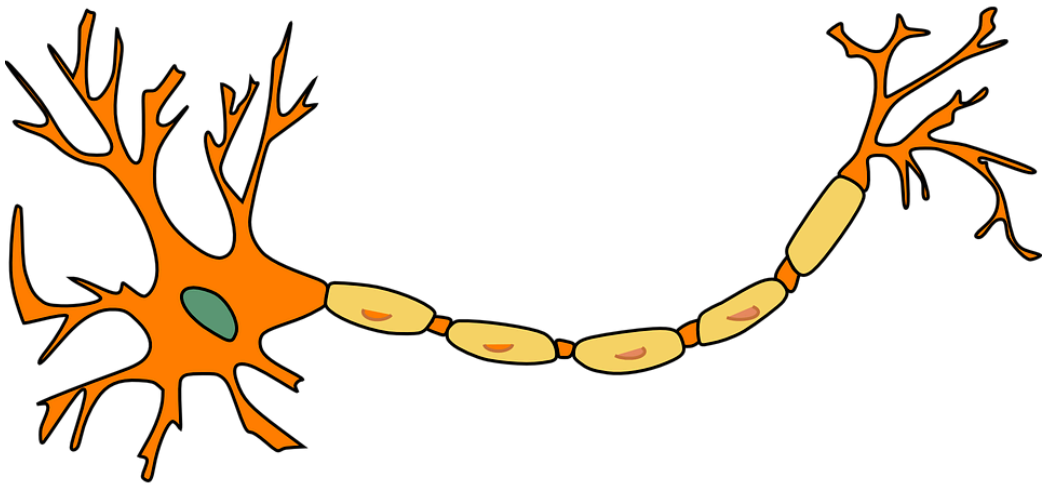
Detecting Features

- Hand-constructed features are time-consuming and labor-intensive
- Often not scalable
- Deep Learning: Construct features automatically from lower level features
- With Artificial Neural Networks



© Alexander Amini and Ava Amini, MIT Introduction to Deep Learning
http://introtodeeplearning.com/slides/6S191_MIT_DeepLearning_L1.pdf

Biological Inspiration



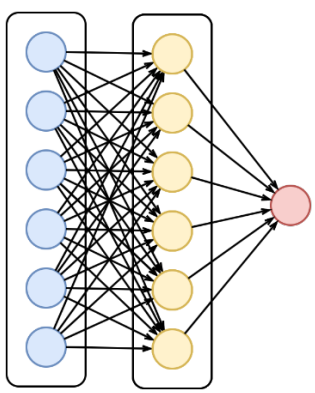
- How?
- Intuition inspired by the biology of the human brain
- Human Neuron
  - Electrically excitable cells
  - Connected to other cells via synapses

https://en.wikipedia.org/wiki/Artificial_neural_network

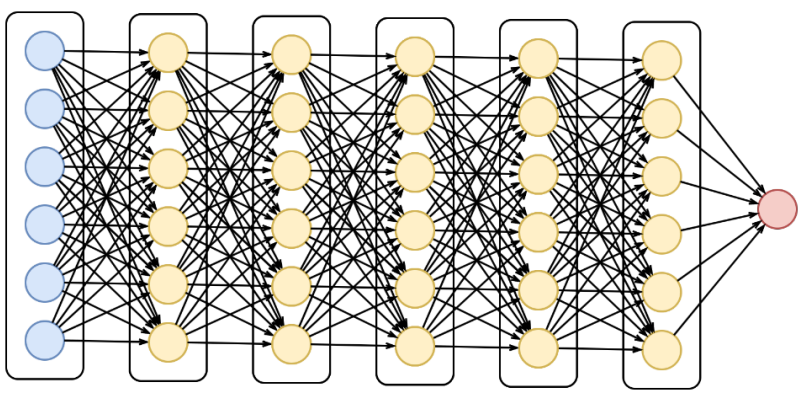
"Deep" Learning

Artificial neural networks are composite functions

"Shallow"



"Deep"



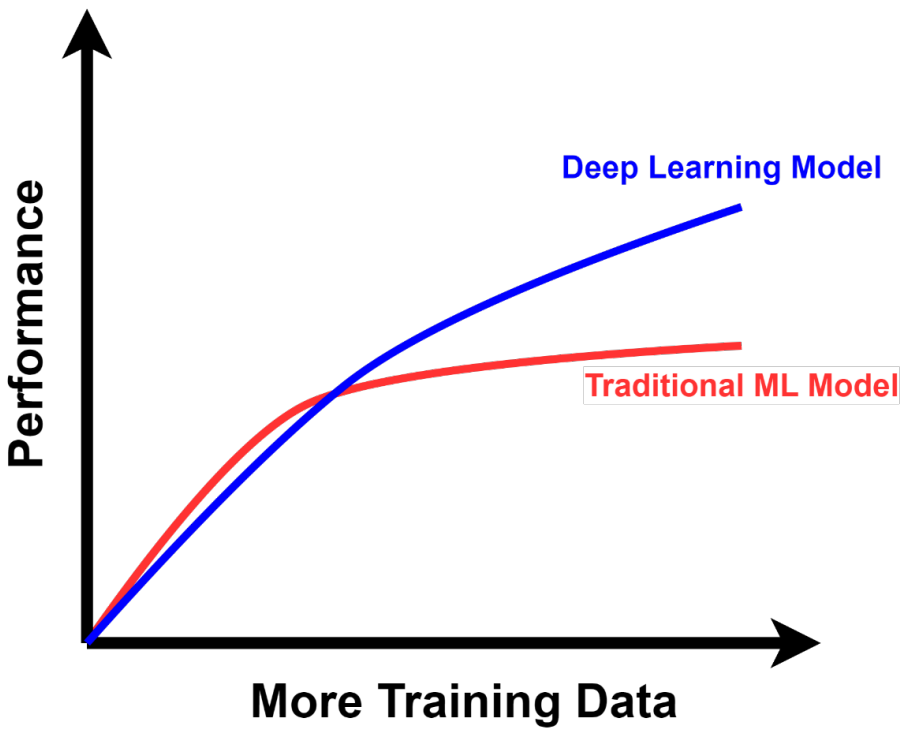
Deep Learning and Representation Learning

- Each layer in the network is a function that
  - Transforms the last representation
  - Into a new representation of the input data
- Many layers ➔ Cascade of layers building on top of each other

> *"Deep learning is a class of machine learning algorithms that: (1) use a cascade of multiple layers of nonlinear processing units for feature extraction and transformation. Each successive layer uses the output from the previous layer as input, (2) learn multiple levels of representations that correspond to different levels of abstraction; the levels form a hierarchy of concepts."*

Zhang, W. J., et al. "On definition of deep learning." *2018 World automation congress (WAC).* IEEE, 2018.

The main claim of Deep Learning



Idea: https://machinelearningmastery.com/what-is-deep-learning/

Application driven

- Deep Learning is much driven by practical applications
- Practical success is often valued more than theory and rigor
- More engineering discipline than science?

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1.4 A Short History of Deep Learning



The early years

- **1943: McCulloch and Pitts**
  - Early model for neural activation
  - Linear threshold neuron (binary): $f_{\boldsymbol{w}}(\boldsymbol{x}) = \begin{cases} +1 \text{ if } \boldsymbol{w}^T\boldsymbol{x} \geq 0 \\ -1 \text{ otherwise} \end{cases}$
- **1958–1962: Rosenblatt's Perceptron**
  - First algorithm and implementation to train single linear threshold neuron
  - Optimization of the perceptron criterion (sum over misclassified examples $\mathcal{M}$):
    $$E(\boldsymbol{w}) = -\sum_{i \in \mathcal{M}} y_i \boldsymbol{w}^T \boldsymbol{x}_i$$
- **1979: Fukushima's Neocognitron**
  - Multilayer network with different types of cells
  - Inspiration for modern CNNs

The viewpoint of the 50s…

https://www.youtube.com/watch?v=cNxadbrN_aI

Minsky and Papert: Perceptrons

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- 1969
- Several negative results
- Famously:
  The XOR-problem
  Simple perceptrons cannot learn many functions

- In the following decade: The (first) “AI Winter”

- Symbolic AI dominates the 70s (and 80s)

Minsky and Papert: Perceptrons: An introduction to computational geometry. MIT Press, 1969.

Summarized from: A. Geiger Deep Learning; Lecture 01 - Introduction

Backpropagation

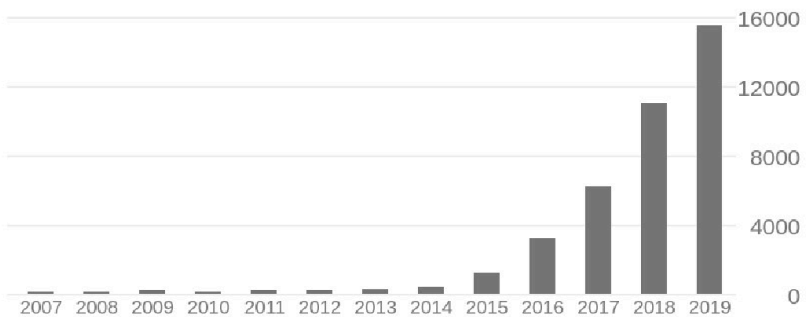
- 1986
- Efficient calculation of gradients in a deep network w.r.t. the network weights
- Enables application of gradient-based learning to deep networks
- Underlying idea is older (Kelley 1960, Linnainmaa 1970, Werbos 1982), but first empirical success for neural networks in 1986
- Remains main workhorse today

Rumelhart, Hinton and Williams: Learning representations by back-propagating errors. Nature, 1986.

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Long Short-Term Memory

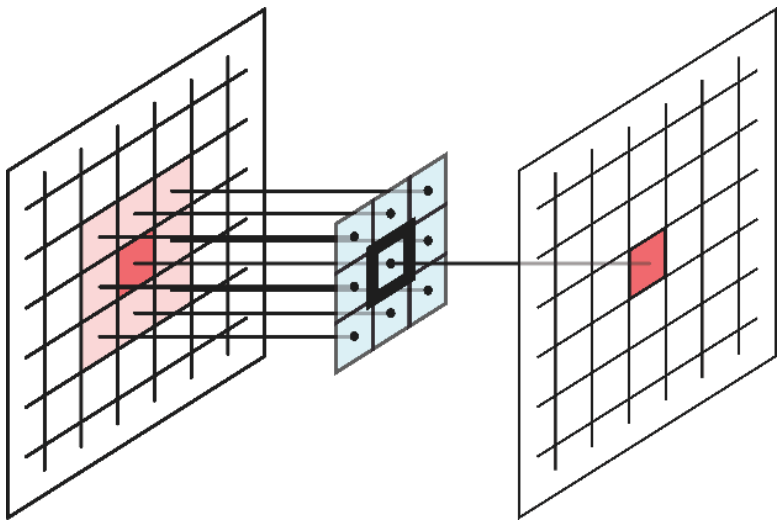
- 1997
- In 1991, Hochreiter demonstrated the problem of vanishing/exploding gradients in his diploma thesis
- Led to the development of long short-term memory for sequence modeling
- Uses feedback and forget/keep gates



Scholar.google.com citations of the LSTM paper

Convolutional Neural Networks

- 1998
- Similar to Neocognitron, but trained end-to-end using backpropagation
- Implements spatial invariance via convolutions and max-pooling
- Weight sharing reduces parameters
- Tanh/Softmax activations
- Good results on MNIST
- But did not scale up (yet)



From: “Deep Learning - A Visual Approach” by Andrew Glassner

Term Deep Learning

- 2006
- Hinton et al.: Training **Deep** Belief **Networks**
- Deep Learning vs Multilayer neural networks:
  - Intentional re-branding
  - No single new invention
  - Many small improvements and scaling to large data

Breakthrough

- ImageNet: 2009
  - Large benchmark dataset: > 14 million images, > 20,000 categories
  - ImageNet challenge (ILSVRC): 1.2 million training images, 1000 categories
- GPU Computation:
  - 2009 / 2010
  - Raina and Ciresan train Deep Neural Networks with GPUs
  - Massive parallelism enables larger-scale computations
- AlexNet: 2012
  - Deep CNN trained on GPUs wins ILSVRC
  - Top-5 error 15.3% vs. 26.2% for the runner-up

Krizhevsky, Sutskever, Hinton. ImageNet classification with deep convolutional neural networks. NIPS, 2012.

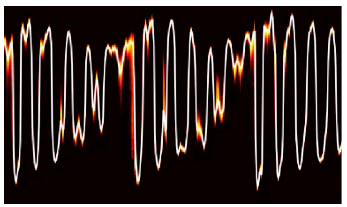
Pretraining and Word Embeddings

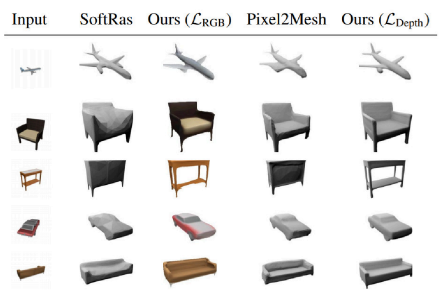
- 2013 / 2014
- Word2Vec, GloVe
- Word representations can be trained (self-supervised) on a large general text corpus
- These pretrained representations can then be reused for specific tasks with smaller training datasets (transfer learning)
- Precursor of today's pretrained *foundation models* (term coined in 2021)

Deep Learning – Towards the Hype

- Since 2014 Deep Learning starts to dominate Machine Learning
  - First images
  - Then texts
  - Networks / other complex data

- New techniques and applications (see above)
  - 2016: WaveNet: Generates speech & music
  - 2016: Style Transfer
  - 2016: AlphaGo
  - 2016 onwards: 3D Deep Learning





Oord et al.: WaveNet: A Generative Model for Raw Audio. Arxiv, 2016.
Niemeyer, Mescheder, Oechsle, Geiger: Differentiable Volumetric Rendering: Learning Implicit 3D Representations without 3D Supervision. CVPR, 2020.

Adapted from: A. Geiger Deep Learning; Lecture 01 - Introduction

Transformers and BERT

- 2017 / 2018
- Transformers: Attention replaces recurrence and convolutions
- BERT: Pre-training of language models on unlabeled text
- GLUE: Superhuman performance on some language understanding tasks
- (paraphrase, question answering, ..)
- But: Computers still fail in dialogue

Age of Large Language Models

- 2020
- GPT-3
- Language model by OpenAI
- 175 Billion parameters
- Text-in / text-out interface
- Many use cases: coding, poetry, blogging, news articles, chatbots
- Controversial discussions

Instruction Tuning & RLHF

- Problem: a pretrained language model only continues text. It does not follow instructions or behave like an assistant
- Solution (InstructGPT, 2022): fine-tune with human feedback in three steps

1 Supervised fine-tuning

Humans write example answers to prompts; the model is trained to imitate them

→

2 Reward model

Humans rank several model answers; a second model learns to predict these preferences

→

3 Reinforcement learning

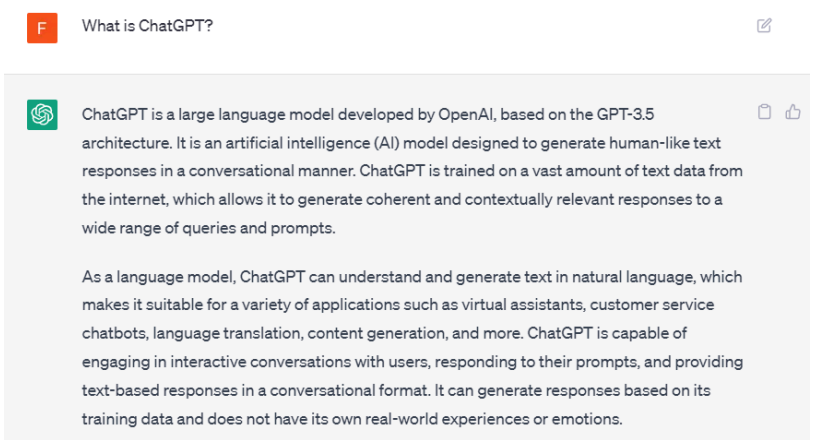
The language model is optimized (PPO) to produce answers with high predicted reward

- Labelers preferred the 1.3B-parameter InstructGPT over the 175B-parameter GPT-3

Ouyang et al.: Training language models to follow instructions with human feedback. NeurIPS, 2022

ChatGPT

- November 2022
- First large language model with deep societal impact
- Estimated 100 million users within two months



https://chat.openai.com/

Since ChatGPT

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2023 Competition

- GPT-4: text *and* image input
- Competitors: Claude, Gemini
- Open-weight models: Llama, Mistral
- LLMs become a product category

2024 Multimodality

- GPT-4o: real-time speech, images, text
- Sora: text-to-video
- o1: “reasoning” by thinking longer before answering
- Nobel Prizes for AI research

2025 Reasoning & Agents

- DeepSeek-R1: open reasoning model, trained with reinforcement learning
- Agents: models use tools, browse the web, write and run code
- Video with sound (Veo 3)

Trends: RL post-training & test-time compute · from chatting to acting · open models close the gap

2026: Where Are We Now?

- **Reasoning is standard:** models “think” (chain of thought) before answering
  → more computation at answer time gives better results in math and science
- **Agents in production:** coding agents, computer use, multi-step workflows
  → the length of tasks AI agents can complete has doubled roughly every 7 months
- **Open-weight models** (e.g., DeepSeek, Qwen) are close to the frontier at much lower cost
- **Pace:** new frontier models every few weeks; massive investments in compute

International AI Safety Report 2026, arXiv:2602.21012
Kwa et al. (METR): Measuring AI Ability to Complete Long Tasks, 2025

Image Generation from Prompts

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- **2021:** DALL-E, CLIP: connecting text and images
- **2022:** Breakthrough of diffusion models
  - DALL-E 2, Midjourney, Stable Diffusion (open weights)
- **2025:** Image generation inside multimodal LLMs (GPT-4o, Gemini “Nano Banana”)
  - Readable text in images, editing by conversation
- **Today:** main model families
  - GPT Image (OpenAI), Nano Banana (Google), Midjourney, Seedream (ByteDance)
  - Open weights: FLUX (Black Forest Labs), Qwen-Image
- Video generation follows the same path: Sora, Veo, Kling, …

Rankings change quickly, see e.g. lmarena.ai/leaderboard

Recognition

- **2018 Turing Award:** Yoshua Bengio, Geoffrey Hinton, Yann LeCun
  “for conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing”
- **2024 Nobel Prize in Physics:** John Hopfield, Geoffrey Hinton
  “for foundational discoveries and invention that enable machine learning with artificial neural networks”
- **2024 Nobel Prize in Chemistry:** David Baker (computational protein design);
  Demis Hassabis, John Jumper (protein structure prediction: AlphaFold)
- **2024 Turing Award:** Andrew Barto, Richard Sutton (reinforcement learning)

https://www.nobelprize.org/all-nobel-prizes-2024/ · https://awards.acm.org/turing

Current Challenges

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Trustworthiness

- Hallucinations
- Interpretability: what happens inside the model?
- Robustness: adversarial attacks, jailbreaks, prompt injection
- Bias and fairness

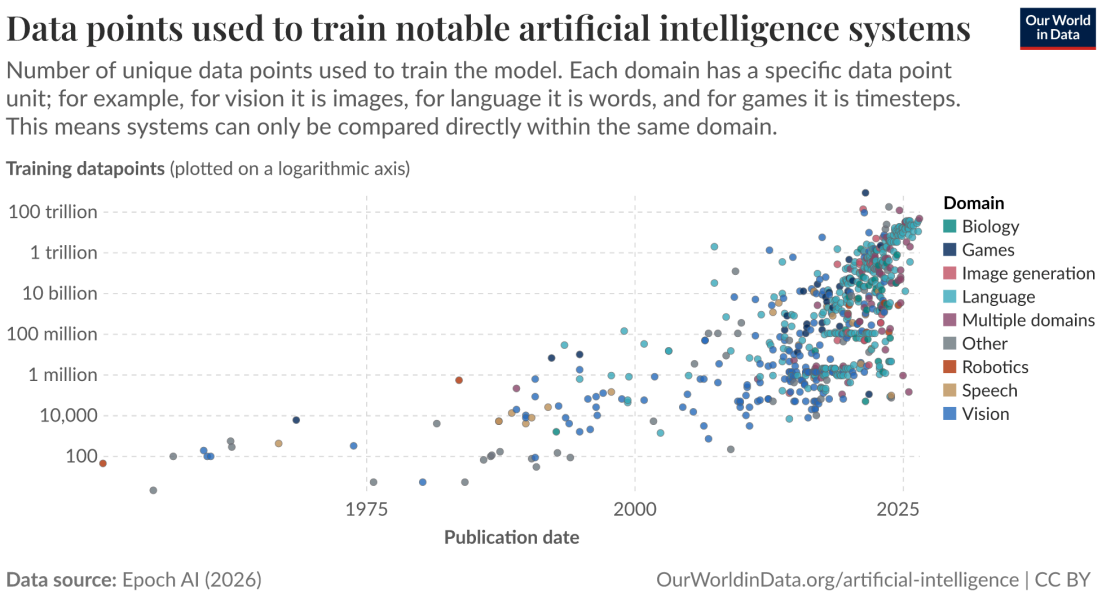
Safety & Society

- Alignment: do models do what we intend?
- Autonomous agents acting in the real world
- Deepfakes and misinformation
- Impact on education and jobs

Resources & Rules

- Energy and compute cost
- Running out of training data, synthetic data
- Copyright and privacy of training data
- Regulation: EU AI Act

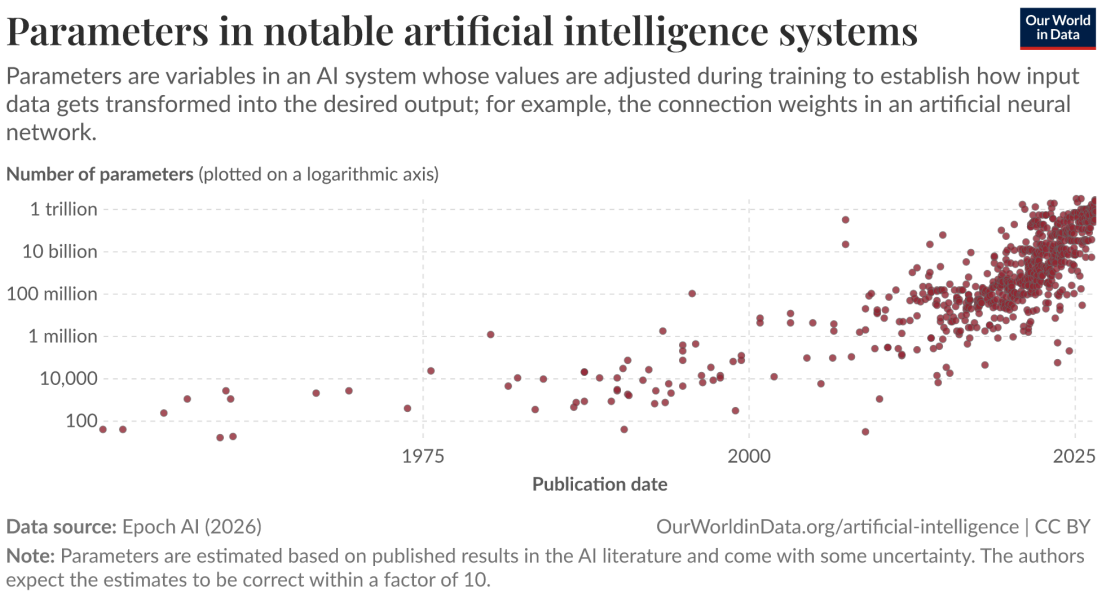
Exponential Growth in Deep Learning



Growing dataset sizes

Data: Epoch AI (2026), chart: Our World in Data, CC BY

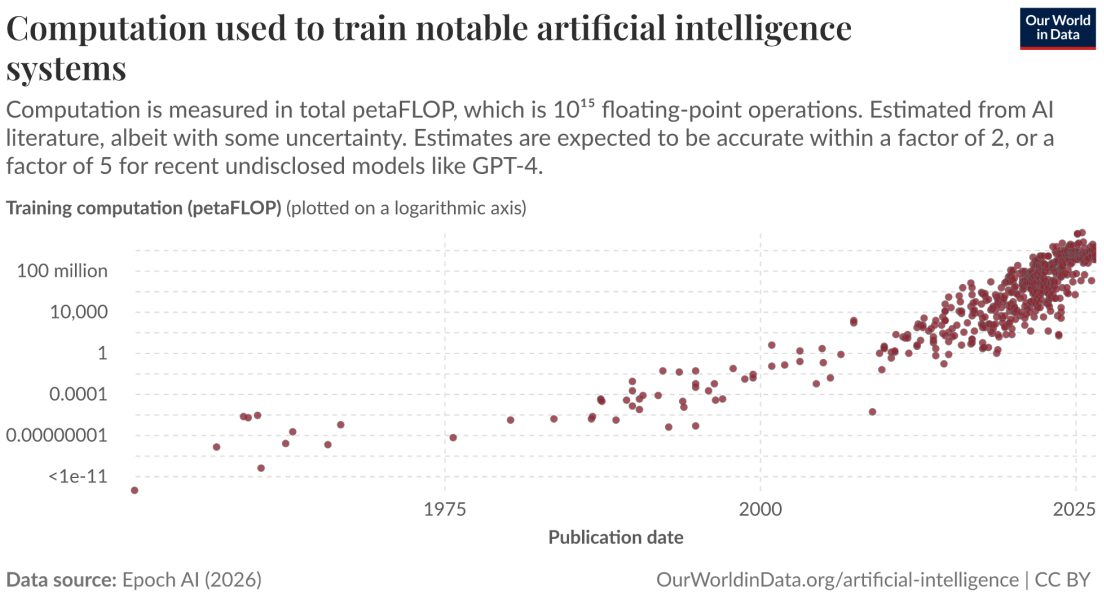
Exponential Growth in Deep Learning



Growing model sizes

Data: Epoch AI (2026), chart: Our World in Data, CC BY

Exponential Growth in Deep Learning



Growing training compute

Data: Epoch AI (2026), chart: Our World in Data, CC BY